A First Course in Abstract Algebra

Section 0: Sets and Relations

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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0 Sets and Relations

Starting point. Read the defining condition before calculating. An equality of sets requires both inclusions; an equivalence relation requires all three properties.

Exercise 0.1: Real square roots

The equation $x^2 = 3$ has two real solutions, $x = \sqrt{3}$ and $x = -\sqrt{3}$. Both square to 3, and factoring $x^2 - 3 = (x - \sqrt{3})(x + \sqrt{3})$ shows there are no others. The set is $\{-\sqrt{3}, \sqrt{3}\}$.

Exercise 0.2: Integer square roots

An integer whose square is 3 would have absolute value strictly between 1 and 2, which is impossible. Thus the set is $\varnothing$. Notice that changing the domain from $\mathbb{R}$ to $\mathbb{Z}$ changes the answer.

Exercise 0.3: Divisors of sixty

The condition means that m divides 60. Since $60 = 2^2 \cdot 3 \cdot 5$, its positive divisors are $1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60$. Every negative of such a number is also a divisor, using a negative quotient. Thus the set is

$$\{\pm 1, \pm 2, \pm 3, \pm 4, \pm 5, \pm 6, \pm 10, \pm 12, \pm 15, \pm 20, \pm 30, \pm 60\}.$$

Zero is excluded because $0n$ cannot equal 60.

Exercise 0.4: An integer inequality

Solve $m^2 - m < 115$ by checking where the upward-opening quadratic crosses zero. Its roots are $(1 \pm \sqrt{461})/2$, approximately -10.235 and 11.235. Hence the integers strictly between them are

$$\{-10, -9, -8, -7, -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}.$$

At the endpoints $m = -10, 11$ the value is 110; at the next integers $-11, 12$ it is 132.

Exercise 0.5: The word large

This is not well defined: no cutoff for "large" is specified, so membership cannot be decided unambiguously. For example, $\{n \in \mathbb{Z}^+ : n > 1000\}$ would be a well-defined replacement, but its cutoff is additional information, not part of the question.

Exercise 0.6: Negative integer squares

Every integer square is nonnegative. Thus no integer satisfies $n^2 < 0$, and the specified object is the well-defined empty set $\varnothing$.

Exercise 0.7: An interval for a cube

Since $3^3 = 27$, $4^3 = 64$, and cubing is strictly increasing, there is no integer with $39 < n^3 < 57$. The set is well defined and equals $\varnothing$.

Exercise 0.8: Almost an integer

"Almost" gives no numerical tolerance. Therefore this description does not define a set. A precise replacement could be $\{q \in \mathbb{Q} : |q - m| < 1/100 \text{ for some } m \in \mathbb{Z}\}$; other tolerances give other sets.

Exercise 0.9: A large denominator

This is a well-defined description of all of $\mathbb{Q}$. If $q = a/b$ with $b > 0$, then $q = (ka)/(kb)$ for every positive integer k. Choose k so that $kb > 100$. The question does not require the fraction to be in lowest terms.

Exercise 0.10: A small positive denominator

The allowed denominators are $1, 2, 3$. Consequently the set is

$$\{a/b : a \in \mathbb{Z},\ b \in \{1, 2, 3\}\} = \mathbb{Z} \cup \tfrac{1}{2}\mathbb{Z} \cup \tfrac{1}{3}\mathbb{Z}.$$

This is well defined. For example $1/2$ is included and $1/4$ is not: $1/4 = a/b$ would force $b = 4a$, impossible for $b = 1, 2, 3$.

Exercise 0.11: A Cartesian product

Choose a first coordinate from $\{a, b, c\}$ and independently a second from $\{1, 2, c\}$. The result is

$$\{(a, 1), (a, 2), (a, c), (b, 1), (b, 2), (b, c), (c, 1), (c, 2), (c, c)\}.$$

There are $3 \cdot 3 = 9$ ordered pairs; an ordered pair is not an unordered set.

Exercise 0.12: Functions, injections, and surjections

A function must assign exactly one value to each of $1, 2, 3$; it is injective when those values are distinct, and onto when all of $2, 4, 6$ occur.

(a) A function, but not injective since $1, 2$ both map to 4, and not onto since 2 is absent.

(b) A function, but not injective since $1, 3$ both map to 4, and not onto since 2 is absent.

(c) Not a function: 1 has three different outputs, while $2, 3$ have none.

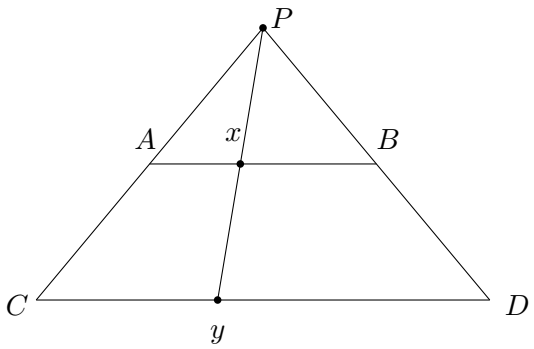
(d) A function and a bijection: $1 \mapsto 6$, $2 \mapsto 2$, $3 \mapsto 4$ uses each target exactly once.

(e) A function, but not injective since all inputs map to 6, and not onto since $2, 4$ are absent.

(f) Not a function: 2 has outputs 6 and 4, and 3 has no output.

Exercise 0.13: Pairing two segments

Draw the line through P and x, and take $y = Px \cap CD$. Projection from P sends A to C, B to D, and each intermediate point to exactly one intermediate point. Conversely the line from P through any point on CD meets AB once. Thus this pairing is a bijection.



Exercise 0.14: Bijections of closed intervals

(a) $f(x) = 2x$ maps $[0, 1]$ onto $[0, 2]$; its inverse is $y/2$.

(b) $f(x) = 10x - 5$ maps 1 to 5 and 3 to 25; its inverse is $(y + 5)/10$.

(c) For $a < b$ and $c < d$, put $f(x) = c + (d - c)(x - a)/(b - a)$. The positive slope makes it injective; for any $y \in [c, d]$, the inverse $x = a + (b - a)(y - c)/(d - c)$ lies in $[a, b]$. Thus it is onto as well.

Exercise 0.15: An open interval and the real line

Use $f(x) = \tan(\pi(x - 1/2))$ for $0 < x < 1$. Its argument ranges through $(-\pi/2, \pi/2)$, where tangent is strictly increasing and has limits $-\infty, +\infty$. Explicitly $f^{-1}(y) = 1/2 + (\arctan y)/\pi$ lies in $(0, 1)$ for every real y. This proves both injectivity and surjectivity.

Exercise 0.16: Four power sets

The empty subset must be included, even when the original set is empty.

(a) $\mathcal{P}(\varnothing) = \{\varnothing\}$ has 1 element.

(b) $\mathcal{P}(\{a\}) = \{\varnothing, \{a\}\}$ has 2 elements.

(c) $\mathcal{P}(\{a, b\}) = \{\varnothing, \{a\}, \{b\}, \{a, b\}\}$ has 4 elements.

(d) $\mathcal{P}(\{a, b, c\}) = \{\varnothing, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$ has 8 elements.

Exercise 0.17: Counting subsets

If $|A| = s$, then $|\mathcal{P}(A)| = 2^s$. Label the elements $a_1, \ldots, a_s$. To specify a subset, decide independently for each a_i whether to include it. There are two choices at each of s positions, hence 2^s subsets. This

also covers $s = 0$, when there is just the empty subset.

Exercise 0.18: Subsets and binary-valued functions

For $f : A \to \{0, 1\}$ define $T(f) = \{a \in A : f(a) = 1\}$. If $T(f) = T(g)$, then for every a either both functions equal 1 or both equal 0, so $f = g$. Given a subset $U \subseteq A$, its characteristic function χ_U, equal to 1 on U and 0 elsewhere, satisfies $T(\chi_U) = U$. Thus T is a bijection, for finite or infinite A.

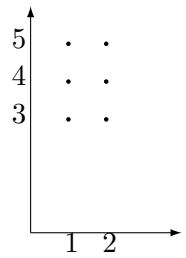
Exercise 0.19: Cantor diagonalization

Suppose $f : A \to \mathcal{P}(A)$ were onto. Define $D = \{a \in A : a \notin f(a)\}$. If $D = f(d)$, then $d \in D$ holds exactly when $d \notin f(d) = D$, a contradiction. Hence no such surjection exists, while $a \mapsto \{a\}$ is injective. Repeated power sets therefore give strictly increasing infinite cardinalities starting with $\mathbb{Z}$. A set U containing everything would contain every subset of itself, so $\mathcal{P}(U) \subseteq U$; this contradicts the same diagonal argument (or forms the contradictory set $\{x \in U : x \notin x\}$). Thus a universal set is not acceptable in ordinary set theory.

Exercise 0.20: Cardinal sums and products

(a) Since A, B are disjoint, their union has the five elements $1, 2, 3, 4, 5$. Similarly adjoining three new objects to $\mathbb{Z}^+$ still permits one infinite list, so $3 + \aleph_0 = \aleph_0$. Two disjoint copies of $\mathbb{Z}^+$ interleave, for example $(n, 0) \mapsto 2n - 1$, $(n, 1) \mapsto 2n$; hence $\aleph_0 + \aleph_0 = \aleph_0$.

(b) $A \times B$ has the six plotted positions $(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5)$. An infinite positive-integer grid can be listed diagonal by diagonal, so $\aleph_0\aleph_0 = \aleph_0$.



Exercise 0.21: Infinite strings of digits

There are $10^2 = 100$ two-place strings and $10^5 = 100000$ five-place strings. Each fixed-length string gives a distinct number in $[0, 1)$, allowing leading and trailing zeros. Infinite decimal strings give $10^{\aleph_0} = |\mathbb{R}|$: every point of $[0, 1)$ has a decimal expansion, and the ambiguity of terminating decimals does not change this cardinality. More explicitly $2^{\aleph_0} \le 10^{\aleph_0} \le (2^4)^{\aleph_0} = 2^{\aleph_0}$, and binary strings inject into $[0, 1]$ as ternary expansions using only digits $0, 2$. Since $2 \le 12 \le 2^4$, the same argument gives $12^{\aleph_0} = 2^{\aleph_0} = |\mathbb{R}|$.

Exercise 0.22: An increasing cardinal sequence

Cantor's theorem says $|X| < 2^{|X|}$. Since $|\mathbb{R}| = 2^{\aleph_0}$, the requested continuation is

$$\aleph_0, \quad |\mathbb{R}|, \quad 2^{|\mathbb{R}|}, \quad 2^{2^{|\mathbb{R}|}}, \quad 2^{2^{2^{|\mathbb{R}|}}}.$$

Each next term is the cardinality of the power set of a set having the preceding cardinality.

Exercise 0.23: Counting partitions of 1 element

The number is $B_1 = 1$, where B_n denotes the number of partitions of an n-element set. The sole partition is $\{\{a\}\}$.

Exercise 0.24: Counting partitions of 2 elements

The number is $B_2 = 2$, where B_n denotes the number of partitions of an n-element set. They are $\{\{a, b\}\}$ and $\{\{a\}, \{b\}\}$.

Exercise 0.25: Counting partitions of 3 elements

The number is $B_3 = 5$, where B_n denotes the number of partitions of an n-element set. The cell-size patterns are 3 (one partition), $2 + 1$ (three choices for the singleton), and $1 + 1 + 1$ (one partition), giving $1 + 3 + 1 = 5$.

Exercise 0.26: Counting partitions of 4 elements

The number is $B_4 = 15$, where B_n denotes the number of partitions of an n-element set. The patterns $4, 3 + 1, 2 + 2, 2 + 1 + 1, 1 + 1 + 1 + 1$ contribute respectively $1, 4, 3, 6, 1$: for two pairs divide the six choices of a first pair by two. Their sum is 15.

Exercise 0.27: Counting partitions of 5 elements

The number is $B_5 = 52$, where B_n denotes the number of partitions of an n-element set. Choose the cell containing a distinguished element. If exactly k of the other four elements lie outside that cell, choose them in $\binom{4}{k}$ ways and partition them in B_k ways. With $B_0 = 1$, this gives $B_5 = \sum_{k=0}^{4} \binom{4}{k} B_k = 1 + 4 + 12 + 20 + 15 = 52$.

Exercise 0.28: A partition gives an equivalence relation

Every element x belongs to the same cell as itself, proving reflexivity. For transitivity, suppose x, y share a cell C and y, z share a cell D. A partition puts y into exactly one cell, so $C = D$. Therefore x, z share a cell. Together with the symmetry already discussed in the text, all three properties hold.

Exercise 0.29: A strictly positive product

This relation is not reflexive on $\mathbb{Z}$: $0 \cdot 0 = 0$ is not strictly positive. Thus it is not an equivalence relation on the stated domain. On $\mathbb{Z} \setminus \{0\}$ it would have the two classes of positive and negative integers, but removing zero changes the question.

Exercise 0.30: The order relation

The relation $x \geq y$ is reflexive and transitive, but is not symmetric: $1 \geq 0$ while $0 \not\geq 1$. Hence it is not an equivalence relation, and no equivalence-class partition is associated to it.

Exercise 0.31: Equal absolute values

Equality of absolute values is reflexive, symmetric, and transitive because equality itself has those properties. The class of x is $\{x, -x\}$, with the special singleton class $\{0\}$. These classes partition $\mathbb{R}$.

Exercise 0.32: Distance at most three

Reflexivity holds since $|x - x| = 0$, and symmetry holds since $|x - y| = |y - x|$. Transitivity fails: $|0 - 3| = 3$ and $|3 - 6| = 3$, but $|0 - 6| = 6 > 3$. Thus this is not an equivalence relation.

Exercise 0.33: Equal numbers of decimal digits

The number of decimal digits is a function from $\mathbb{Z}^+$ to $\mathbb{Z}^+$. Equality of its values is an equivalence relation. The classes are

$$\{1, \ldots, 9\}, \quad \{10, \ldots, 99\}, \quad \{100, \ldots, 999\}, \quad \ldots,$$

or $\{10^{k-1}, \ldots, 10^k - 1\}$ for $k \geq 1$.

Exercise 0.34: Equal final digits

Two positive integers have the same final digit exactly when they have the same remainder modulo 10. Equality of remainders is reflexive, symmetric, and transitive. The ten classes are $\{10, 20, 30, \ldots\}$ and $\{r, r + 10, r + 20, \ldots\}$ for $r = 1, \ldots, 9$.

Exercise 0.35: Positive residue classes

(a) For $n = 2$: $\{2, 4, 6, \ldots\}$ and $\{1, 3, 5, \ldots\}$.

(b) For $n = 3$: $\{3, 6, 9, \ldots\}$, $\{1, 4, 7, \ldots\}$, $\{2, 5, 8, \ldots\}$.

(c) For $n = 5$: $\{5, 10, 15, \ldots\}$, $\{1, 6, 11, \ldots\}$, $\{2, 7, 12, \ldots\}$, $\{3, 8, 13, \ldots\}$, $\{4, 9, 14, \ldots\}$.

The class beginning with n is the zero-remainder class, even though zero is absent from $\mathbb{Z}^+$.

Exercise 0.36: Congruence on all integers

(a) $r - r = n \cdot 0$ proves reflexivity. If $r - s = nq$, then $s - r = n(-q)$ proves symmetry. If $r - s = nq$ and $s - t = np$, then $r - t = n(q + p)$ proves transitivity.

(b) Write $r = nq + a$, $s = np + b$ with $0 \leq a, b < n$. Their difference is divisible by n exactly when $a - b$ is; since $|a - b| < n$, this happens exactly when $a = b$. Thus the restriction is equality of remainders.

(c) For $n = 2$ the classes are $\{\ldots, -4, -2, 0, 2, 4, \ldots\}$ and $\{\ldots, -3, -1, 1, 3, \ldots\}$. For $n = 3$ they are $\{\ldots, -6, -3, 0, 3, 6, \ldots\}$, $\{\ldots, -5, -2, 1, 4, 7, \ldots\}$, $\{\ldots, -4, -1, 2, 5, 8, \ldots\}$. For $n = 5$ they are $\{\ldots, -10, -5, 0, 5, 10, \ldots\}$, $\{\ldots, -9, -4, 1, 6, 11, \ldots\}$, $\{\ldots, -8, -3, 2, 7, 12, \ldots\}$, $\{\ldots, -7, -2, 3, 8, 13, \ldots\}$, and $\{\ldots, -6, -1, 4, 9, 14, \ldots\}$.

Exercise 0.37: What injectivity actually says

Every function already sends one input to one output. Injectivity adds that *two different inputs must produce two different outputs*: $a \neq b \Rightarrow f(a) \neq f(b)$. Calling this "two-to-two" would emphasize comparing two distinct inputs rather than describing the image of just one input. The logically equivalent test is $f(a) = f(b) \Rightarrow a = b$; no assumption of being onto is involved.